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STEP 1 — AWS CONSOLE: CREATE EBS VOLUME

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EC2

- Elastic Block Store
- Volumes
- Create volume

Select:

Volume type : gp3

Size : 60 GiB

Availability Zone: **SAME AZ as EC2 server**

Encryption : Enabled

Click: Create volume

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STEP 2 — AWS CONSOLE: ATTACH EBS VOLUME

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EC2

- Volumes
- Select the new 60 GiB volume
- Actions
- Attach volume

Instance:

ip-10-20-3-149

Device name:

/dev/sdf

Click: Attach volume

NOTE:

AWS may show /dev/sdf, but Linux may show it as /dev/nvme1n1. This is normal.

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STEP 3 — CONNECT TO THE SERVER

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Connect to:

ip-10-20-3-149

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STEP 4 — CHECK THE NEW DISK

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lsblk

EXPECTED:

```
nvme0n1    50G disk
├─nvme0n1p1
├─nvme0n1p2    /boot/efi
└─nvme0n1p3    /

nvme1n1    60G disk    <-- NEW EBS
```

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STEP 5 — VERIFY NEW DISK IS EMPTY

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sudo lsblk -f

EXPECTED:

```
nvme0n1
├─nvme0n1p1
├─nvme0n1p2 vfat    /boot/efi
└─nvme0n1p3 btrfs   /

nvme1n1    60G
```

IMPORTANT:

For nvme1n1:

FSTYPE = blank

MOUNTPOINT = blank

If nvme1n1 already has a filesystem/data:

STOP.

DO NOT RUN mkfs.

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STEP 6 — FORMAT NEW DISK
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sudo mkfs.xfs /dev/nvme1n1

EXPECTED:

meta-data=/dev/nvme1n1
data =...
naming =version 2
log =internal log
realtime =none

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STEP 7 — CREATE MOUNT DIRECTORY
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sudo mkdir -p /usr/sap

EXPECTED:

No output = SUCCESS

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STEP 8 — MOUNT THE DISK
=====

sudo mount /dev/nvme1n1 /usr/sap

EXPECTED:

No output = SUCCESS

=====
STEP 9 — VERIFY MOUNT

```
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```

df -hT /usr/sap

EXPECTED:

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
/dev/nvme1n1	xfs	60G	1.2G	59G	2%	/usr/sap

```
=====
```

STEP 10 — GET UUID

```
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```

sudo blkid /dev/nvme1n1

EXPECTED:

/dev/nvme1n1: UUID="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx" TYPE="xfs"

COPY THIS UUID.

```
=====
```

STEP 11 — BACKUP FSTAB

```
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```

sudo cp /etc/fstab /etc/fstab.backup-\$(date +%F-%H%M%S)

EXPECTED:

No output = SUCCESS

```
=====
```

STEP 12 — EDIT FSTAB

```
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```

sudo vi /etc/fstab

ADD:

UUID=YOUR-UUID-HERE /usr/sap xfs defaults,nofail 0 0

Example:

```
UUID=edf41c1c-6c5f-4937-80ec-e9ab1b1ca92b /usr/sap xfs defaults,nofail 0 0
```

Save:

```
ESC  
:wq  
ENTER
```

```
=====
```

STEP 13 — TEST FSTAB

```
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```

sudo umount /usr/sap

EXPECTED:

No output = SUCCESS

sudo mount -a

EXPECTED:

No output = SUCCESS

```
=====
```

STEP 14 — FINAL VERIFICATION

```
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```

df -hT /usr/sap

EXPECTED:

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
/dev/nvme1n1	xfs	60G	1.2G	59G	2%	/usr/sap

```
=====
```

STEP 15 — VERIFY MOUNT

```
=====
```

findmnt /usr/sap

EXPECTED:

TARGET	SOURCE	FSTYPE
/usr/sap	/dev/nvme1n1	xf

=====

STEP 16 — FINAL CHECK

=====

sudo lsblk -f

EXPECTED:

nvme0n1	...	btrfs	/
nvme1n1	60G	xf	... /usr/sap

=====

RESULT

=====

AWS EBS 60 GB

↓

/dev/nvme1n1

↓

XFS filesystem

↓

/usr/sap

↓

/etc/fstab

↓

Persistent after reboot

IMPORTANT SAFETY RULE:

NEVER blindly run:

sudo mkfs.xfs /dev/nvme1n1

unless lsblk -f confirms that nvme1n1
is the NEW EMPTY 60 GB disk.

Device names can change. Always verify with lsblk first.